

# JUNO LI

Data Scientist | Data Engineer

**Location:** Perth, WA

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## PROFILE SUMMARY

Data professional with more than five years' experience analysing complex datasets, developing statistical models and building reproducible data workflows. Skilled in Python, SQL and R, with practical project experience in Databricks, PySpark, machine learning and automated data pipelines. Experienced in transforming noisy, multi-source data into reliable analytical datasets, investigating data-quality issues and communicating practical insights to technical and non-technical stakeholders.

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## TECHNICAL SKILLS

**Programming & Data:** Python, SQL, R, pandas, NumPy, PostgreSQL, REST APIs, Linux/Bash

**Data Engineering & Platforms:** Databricks, PySpark, SparkSQL, ETL/ELT, data modelling, schema validation, metadata, orchestration, Git, CI/CD fundamentals, introductory dbt

**Analytics & ML:** EDA, regression, Bayesian modelling, forecasting, time-series analysis, uncertainty quantification, scikit-learn, XGBoost, MLflow

**Reporting & Delivery:** Data visualisation, matplotlib, Excel, Power BI and Tableau working knowledge, technical documentation, requirements clarification

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## PROFESSIONAL EXPERIENCE

### The University of Western Australia, Perth

**Postdoctoral Research Associate**

**Apr 2026 – Present**

**Research Assistant, Part-time**

**Oct 2025 – Feb 2026**

- Re-engineer research notebooks and project-specific scripts into modular, maintainable Python and R software, improving reliability through configuration management, automated validation, testing and version control.
- Develop analysis-ready datasets and software outputs supporting collaborative research and publication activities.
- Work with researchers to clarify requirements, investigate data and workflow issues and communicate practical solutions.
- Use AI agents and VS Code-based workflows for refactoring, debugging, testing and documentation, with human review of code and analytical outputs.

**Doctoral Researcher – The University of Western Australia , Perth**

**Mar 2022 – Sep 2025**

- Developed and validated statistical and Bayesian models for complex, noisy observational datasets.
- Built reproducible Python and R workflows covering ingestion, transformation, modelling, validation and visualisation.
- Integrated heterogeneous data from multiple international facilities into consistent, analysis-ready datasets.
- Communicated findings through first-author publications, international conferences and multidisciplinary collaborations.

## SELECTED PROJECTS

### Solar Yield Forecasting MLOps Pipeline – [GitHub](#)

Built an automated workflow that produces and publishes refreshed seven-day solar-yield forecasts from external weather data.

**Tools:** Python, Databricks, PySpark, SparkSQL, XGBoost, MLflow, REST APIs

**Skills:** Forecasting, feature engineering, workflow automation, validation, scheduled inference

### DData Engineering Zoomcamp Capstone Project – [GitLab](#)

Built an end-to-end pipeline that converts raw source data into validated, structured datasets for downstream analysis.

**Tools:** Python, SQL, PostgreSQL, GCP, GitLab

**Skills:** ETL, orchestration, data modelling, schema validation, data quality, CI/CD

### Large-Scale Astronomical Data Pipeline

Processed and indexed more than 20TB of archival imaging data during my PhD, enabling efficient discovery of related observations without repeatedly inspecting large source files.

**Tools:** R, Linux, astronomical image formats, WCS

**Skills:** Metadata engineering, spatial matching, reconciliation, large-scale processing, data-quality investigation

### Spatially Resolved Galaxy Analysis Software – [Architecture](#)

Modernising research workflows into reusable software for broader research use as part of my current postdoctoral role.

**Tools:** Python, R, Git/GitHub, VS Code, AI coding agents

**Skills:** Modular software design, configuration management, testing, validation, documentation, human review

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## ADDITIONAL EXPERIENCE

### Quantitative Developer (Intern) – Mai Capital Ltd, Hong Kong

Jun 2019 – Nov 2019

- Built Python and SQL workflows using broker APIs to monitor portfolio positions, performance and compliance breaches in near real time.
- Applied financial risk metrics and stochastic-process concepts to support strategy evaluation and automated controls.

### Research Assistant – University of Hong Kong, Hong Kong

Dec 2018 – Feb 2022

- Designed and delivered a practical laboratory project involving sequential observations, motion detection and spatial inference.
- Supervised undergraduate and Master's students, translating technical concepts for non-specialist audiences.

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## EDUCATION

PhD in Astrophysics – University of Western Australia (ICRAR)

Completed Feb 2026

MPhil in Astrophysics – The University of Hong Kong, Hong Kong

2016–2019

Bachelor of Science (Hons) – The University of Hong Kong, Hong Kong

2012–2016

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## PROFESSIONAL DEVELOPMENT

Data Engineering Zoomcamp Certificate – DataTalks.Club

Issued Apr 2025

[Google Data Analytics Certificate](#) – Coursera

Issued Sep 2022